



Fig. 3: Circuit for touchscreen control for wheelchair

the change in capacitance at the four corners of the screen.

Surface acoustic wave touchscreen. Surface acoustic wave touchscreen contains two transducers (transmitting and receiving) placed along the X-axis and Y-axis of the monitor's glass plate along with some reflectors. The waves propagate across the glass and reflect back to the sensors.

When the screen is touched, the waves are absorbed and a touch is detected at that point. These reflectors reflect all the electrical signals sent from one transducer to another. This technology provides excellent throughput and image clarity.

Circuit description

Fig. 3 shows the circuit for the touchscreen control for wheelchair. It comprises microcontroller ATmega16 (IC1), motor driver L293D (IC2), regu-

lator 7805 (IC3), resistive touchscreen and a few discrete components.

The ATmega16 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. The AVR core combines a rich instruction set with 32 general-purpose working registers. All the 32 registers are directly connected to the arithmetic logic unit, allowing two independent registers to be accessed in one single instruction executed in one clock cycle. This architecture is more code-efficient.

The ATmega16 has the following features: 16 kB of in-system programmable Flash program memory, 512 bytes of EEPROM, 1kB SRAM, 32 general-purpose input/output (I/O) lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a byte-oriented two-wire serial interface, an 8-channel 10-bit analogue-to-digital converter (ADC), an SPI serial port, and six software-selectable power-saving modes.

The microcontroller has an 8-channel ADC. Port A of the microcontroller is multiplexed with the 8-channel ADC. Port pins PA0 through PA3 of microcontroller are interfaced with pins 1 through 4 of touchscreen. The touchscreen uses four channels (ADC0 through ADC3). The resistive-type touchscreen used here is quite cheaper than capacitive touchscreens and also easy to use.

When the touchscreen is touched,

its resistance changes depending on the position of the touch. The touch input is in the form of analogue values of the voltage relative to the point of touch. This is processed in microcontroller ATmega16 using the ADC.

The ADC converts the analogue values of the

touchscreen into 10-bit digital equivalent form. This digital data is then processed to find the exact position of the touch and perform the corresponding action like left turn, right turn, forward movement and reverse movement with the help of motor driver L293D. Switch S1 is used for manual reset.

Motor driver L293D is interfaced with the microcontroller's port B. Port pins PB0 and PB1, and PB2 and PB3 of the microcontroller control motors M1 and M2, respectively. Motor drivers are enabled in pairs. When enable input pins 1 and 9 of IC2 are high, the associated drivers are enabled and their output pins 3 and 6 and pins 11 and 14, respectively, are active and in phase with the inputs. With the proper data inputs, each pair of drivers forms an H-bridge reversible drive, suitable for motor applications.

Motors M1 and M2 rotate in forward direction (clockwise) when port pins PB1 and PB3 are high. Motors M1 and M2 rotate in reverse direction (anti-clockwise) when port pins PB0 and PB2 are high.

A 12V DC battery is used to power the circuit. IC 7805 provides regulated 5V supply to the circuit. Capacitor C1 bypasses ripple from the regulated supply. LED1 acts as the power-on indicator and resistor R1 limits the current through LED1.

Construction and testing

An actual-size, single-side PCB for the touchscreen control for wheelchair is

PARTS LIST

Semiconductors:

- IC1 - ATmega16 microcontroller
- IC2 - L293D motor driver
- IC3 - 7805, 5V regulator
- D1-D4 - 1N4148 switching diode
- LED1 - 5mm LED
- Touchscreen module

Resistors (all 1/4-watt, ± 5 per cent carbon):

- R1 - 470-ohm
- R2-R4 - 10-kilo-ohm

Capacitors:

- C1 - 0.1µF ceramic disk

Miscellaneous:

- S1 - Push-to-on tactile switch
- S2 - On/off switch
- M1, M2 - 50-rpm geared DC motor
- BATT. - 12V, 7Ah battery